

DHYAN B

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SUMMARY

Information Science Engineering graduate with strong experience in Full-Stack Development and Generative AI, building and deploying end-to-end intelligent applications. Hands-on with Retrieval-Augmented Generation (RAG), LLM agents with tool calling, embeddings and vector search (FAISS, pgvector), and production-style backends using FastAPI, React and PostgreSQL, containerized with Docker and deployed to the cloud. Consistently sharpening problem-solving skills through rigorous Data Structures and Algorithms (DSA) practice.

SKILLS

Programming Languages: Java, C, Python, JavaScript, TypeScript

Development: HTML, CSS, React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Chrome Extensions (Manifest V3)

AI / GenAI: LLMs (OpenAI, Groq Llama 3.3), RAG, LangChain, AI Agents & Tool Calling, Embeddings, Vector Search (FAISS, pgvector), Hybrid Search, Prompt Engineering

Database: PostgreSQL, Supabase, MySQL, MongoDB, Redis, Firebase, SQLAlchemy

Core Subjects: Computer Networks, Operating Systems, Database Management System, OOPS

Tools and Technologies: Git, GitHub, Docker, AWS, Linux, Render, Railway, uv

EXPERIENCE

Full-Stack Intern

June 2024 – Aug 2024

MathJunior Education

- Developed an Android application that streamed and visualized real-time sensor data using Firebase.
- Processed and displayed real-time sensor data, handling 1,000+ sensor updates per second, integrated with Google Firebase for real-time data management.
- Integrated secure user authentication, enabling 500+ logins via Firebase Authentication.

PROJECTS

PageSense – Agentic RAG Chrome Extension ([GitHub](#)) | FastAPI, LangChain, FAISS, Groq Llama 3.3

July 2026

- Built a Manifest V3 Chrome extension that lets users chat with any webpage in real time, extracting page content and sending it to a FastAPI backend that answers questions using Retrieval-Augmented Generation.
- Implemented the RAG pipeline: 500-character chunking, BAAI/bge-small-en embeddings and FAISS top-3 similarity search, with embedding and retrieval completing in under 0.3 s.
- Designed a custom tool-calling loop that falls back to live DuckDuckGo web search when page context is insufficient, plus per-tab conversational memory using a 5-turn sliding window.
- Reduced LLM API latency from 75 s to 0.6 s by diagnosing an IPv6 DNS resolution delay and forcing IPv4 socket resolution.

AI News Aggregator ([GitHub](#)) | Python, OpenAI, PostgreSQL, SQLAlchemy, FastAPI, Docker, Render

August 2026

- Built an end-to-end GenAI pipeline that scrapes AI content from YouTube channels (video transcripts) and OpenAI / Anthropic blog RSS feeds, converts it to Markdown and stores it in PostgreSQL using SQLAlchemy.
- Developed LLM agents for article summarization, personalized ranking against a user profile and generation of a daily HTML email digest delivered via Gmail SMTP, with tracking to prevent duplicate sends.
- Containerized the application with Docker and deployed it on Render, with a scheduled GitHub Actions workflow running the full scrape → summarize → rank → email pipeline every day.

Document Copilot – RAG over SEC Filings ([GitHub](#)) | FastAPI, React, TypeScript, Supabase, pgvector

September 2026

- Built a full-stack RAG application that lets analysts query SEC 10-K filings in plain English and receive grounded answers with source citations and persistent chat history.
- Implemented a document ingestion pipeline (parsing, chunking, OpenAI embeddings) and hybrid retrieval combining pgvector semantic search with PostgreSQL full-text search.
- Developed a React + TypeScript frontend (Tailwind CSS, shadcn/ui) with Supabase Auth, managed schema with SQLAlchemy and Alembic migrations, and deployed the app on Railway.

ACHIEVEMENTS AND CERTIFICATIONS

- Foundations of Coding Full-Stack – Coursera (2024), offered by Microsoft.
- SAP HackFest 2025, Finalist – Designed and developed AgroLoop, a sustainable web application aimed at empowering farmers through eco-friendly agricultural practices and smart crop planning.

EDUCATION

B.E. in Information Science

National Institute of Engineering, Mysore

Graduated: 2026

CGPA: 8.5 / 10